Can Satellites Replace the Field Plot? A Blind Benchmark of Satellite Forest Carbon and Conformal Crediting with Calibrated Risk

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***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer, and credits are issued with deductions of unknown risk. A right grid computes sealed field ground truth to SNI 7724, IPCC, CDM, and Verra rules; an isolated left grid sees only plot coordinates and dates and estimates carbon from time-matched Sentinel-1/2, Landsat, PALSAR, canopy height, and GEDI lidar. On 476 plots, blind error was 21% on 1 ha plots and 7–9% on project totals outside Indonesia, but near 100% on small Indonesian plots. Counting trees on a 1 m canopy height map found 3% of the stems and 25% of the carbon (62% on the 2026 map). In 1,000 simulated projects the best map found a median 95% of the field carbon; satellite embeddings and a regional bias correction raised the share within ± 25% from 80% to 88.5%, and on 35 unseen Bornean plots the frozen method found 85%. Only 21.5% of Sentinel-2 acquisitions see an Indonesian plot clear. We propose conformal crediting, a lower bound calibrated on other projects with field truth. It met a 10% over-crediting target for five maps within 2.9 points, was invariant to map inflation, and credited 72–84% of the true stock without a field plot, while the VM0047 deduction over-credited 23–29% of projects that had field plots. No satellite estimate of annual growth beat assuming no change.**

***Index Terms*—forest carbon, MRV, remote sensing, cloud cover, conformal prediction, spatial cross-validation, carbon crediting, commit-reveal ledger, distributed processing**

I. Introduction

A forest carbon credit is issued on a number computed by a fixed chain: field crews measure each stem's diameter D and height H and identify its species for wood density ρ, a pantropical allometric equation converts these to aboveground biomass (AGB) [1], and fixed factors convert AGB to carbon and CO_2 [2], [3]. The costly part is the field visit, above all in remote Indonesian forests. Satellite platforms, including our NusaCarbon prototype, promise to replace it, and global biomass maps exist [4], [5], but unverifiable estimates produce over-crediting [6].

How close is a satellite number to the standard formula's at the same place and time? Three traps inflate reported accuracy: *leakage*, training or tuning on plots next to those scored [7]; *temporal mismatch* of field and image dates; and *scale*, comparing 0.05 ha plots with 10 m to 4.6 km pixels. An audit of our own platform found forest figures driven by default constants, scene-level cloud screening, and an unchecked metadata "confidence" grade.

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: public field plots, the standard formula, a carbon density. The **left grid** is free, any satellite data and algorithm, but never sees the right grid's outcomes for the plots it is scored on. We ask: **RQ1** (methodology) how far a blind satellite estimate deviates from the ground truth, by plot size and aggregation; **RQ2** (AI) which inputs and learners close the gap, and whether trees can be counted from space; **RQ3** how often clouds hide an Indonesian plot, and whether uncertainty can be stated honestly; **RQ4** whether a credit can be issued at a stated risk of over-crediting, and growth estimated; and **RQ5** (distributed systems) whether extraction can be sharded and fault tolerant with bit-identical output.

Contributions. (i) A blind, time-matched protocol with a commit-reveal ledger (476 stock, 196 repeat-census plots). (ii) A standards-based right grid with a Monte Carlo uncertainty budget. (iii) A plot-level clear-sky climatology for Indonesia. (iv) A test of missing-data conformal intervals under cloud. (v) *Conformal crediting*, a crediting rule with calibrated over-crediting risk, invariant to map inflation, built on standard conformal machinery [8]. (vi) A tree-by-tree satellite estimator and 1,000 simulated projects. (vii) Fixes for each limit, frozen and tested blind on 35 unseen plots. (viii) A sharded, idempotent

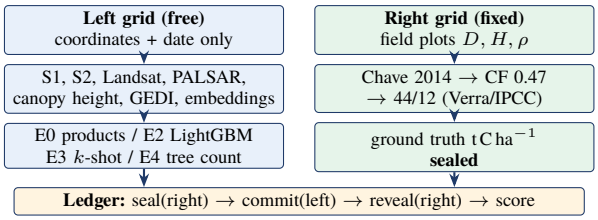


Fig. 1. Blind two-grid design: only id, location, date, and radius cross.

extraction runner.

II. Related Work

Remote sensing. Optical and radar signals saturate over dense canopy. GEDI lidar samples canopy height in 25 m footprints [9]; global maps include ESA CCI Biomass [4], Chloris [5], and 1 m and 10 m canopy height [10], [11]. **Validation and uncertainty.** BIOMASS propagates the uncertainty of Chave's allometry [12]. Conformal prediction gives finite-sample intervals [8], conformalized quantile regression (CQR) adapts their width [13], and Zaffran *et al.* handle missing covariates [14]. Conservative crediting asks that a credit exceed the truth with small probability [15], solved parametrically by Neeff [16].

III. Methodology

A. Right Grid: Ground Truth by the Standards

Data. The stock benchmark has 476 plots in 27 region blocks: 148 HCSA plots in six Indonesian provinces, nested 0.01/0.05 ha [17]; 142 GEO-TREES network plots, pantropical [18]; and 186 Labrière 1 ha units in French Guiana and Gabon [19]. For change, the SAFE census in Sabah gives 25×25 m plots measured stem by stem with day-precision dates from 2010 to 2020 [20]. We follow SNI 7724 [21] (five pools, precision ≤20%), the HCSA toolkit [22], IPCC [3] (Monte Carlo allowed), CDM AR-TOOL14 [23], Verra VM0047 [2] (deduct $U - 10\%$), and Gold Standard [24] (deduct the excess over 20%).

What a plot is. Coordinates only locate a plot; the ground truth is the list of trees inside it. A crew tags every stem above the minimum diameter, measures its diameter D at 1.3 m and its height H with a clinometer, and records its local name, which a botanist resolves to species. Wood density ρ, oven-dry mass per green volume, is looked up by species, genus, or family. HCSA publishes the list: 2,528 stems of 370 species in 89 families, each with D, H, name, and ICRAF ρ [17]; GEO-TREES and Labrière publish plot biomass. None of it reaches the left grid.

Tree, plot, and carbon. A tree's AGB follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ [\mathrm{kg}], \qquad \sigma_{\ln} = 0.357, \quad (1)$$

with H measured (HCSA) or from a local $\ln H$–$\ln D$ model (Labrière) [12]. Tree biomass is expanded by the area in which a stem of its size is counted, and pools and carbon follow IPCC and SNI 7724:

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\, a(D_i)}, \qquad C_j = \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j, \quad (2)$$

with $a(D) = 0.01$ ha for $5 \le D < 15$ cm and 0.05 ha above, $\mathrm{CO_2e}_j = \frac{44}{12} C_j$, CF $= 0.47$, and root-to-shoot $R = 0.37$ (SNI 7724, IPCC 2006) or 0.21 (IPCC 2019). We compare AGB, so CF and R cancel and every relative error holds for carbon and $\mathrm{CO_2e}$. HCSA was recomputed tree by tree; plots with a missing input or an implausible stem (a *Gluta renghas* at $D = 480$ cm) were excluded, never imputed, as were palms, for which Eq. 1 is not made.

Uncertainty and deduction. Plots are grouped into strata h with weights w_h, and the project uncertainty at 90% is $u = t_{0.95}\sqrt{\sum_h w_h^2 s_h^2 / n_h}/\bar{b}$ (AR-TOOL14) [23]. AR-TOOL14 counts sampling error only. Measurement and model errors are large (the allometric residual alone is $\sigma_{\ln} = 0.357$ per tree) and pass through the nonlinear Eq. 1, where summing variances (IPCC Approach 1) is inexact, so we use Approach 2, Monte Carlo: 2,000 draws per tree of diameter, height, wood density, allometric residual, and CF errors with the BIOMASS error models [12]. For change, VM0047 Eq. 28 gives $U = T\sqrt{SE_0^2 + SE_1^2 - 2\rho\, SE_0 SE_1}/\Delta C$, ρ the correlation of the two stock means, and credits are multiplied by $1 - \min(1, \max(0, U - 0.1))$ [2].

B. Left Grid: Inputs and Estimators

From coordinates and the field date t_j only, the left grid reads clear observations within $t_j \pm 90$ days: Sentinel-2 L2A bands and indices, Landsat 5–9, Sentinel-1 backscatter, the annual ALOS PALSAR mosaic, and elevation; through Google Earth Engine also the Meta 1 m and ETH 10 m canopy height maps [10], [11], GEDI L4A biomass and L2A height within 1 km [9], ESA CCI Biomass [4], and Hansen loss [25]; the fixes of Section IV-D add the AlphaEarth embedding (64 features per 10 m pixel) [26] and the 2026 canopy height map [27]. **E0** reads published products as they are. **E2** is LightGBM [28] on $\ln$ AGB with smearing retransformation, trained on all region blocks but the target (leave-one-region-out, LORO). **E3** recalibrates E2 by the ratio on k random plots of the target block. We score relative RMSE (rRMSE), bias, R^2, and the error of the mean of n plots, a project total, with a region-block bootstrap (2,000 draws) over five seeds.

From pixels to carbon. A satellite does not see trees; it records reflected light and radar echoes per pixel (Fig. 2). Optical bands become indices such as the normalized difference vegetation index [29]

$$\mathrm{NDVI} = (R_{\mathrm{NIR}} - R_{\mathrm{red}})/(R_{\mathrm{NIR}} + R_{\mathrm{red}}), \quad (3)$$

with R the reflectance in Sentinel-2 bands B8 (842 nm) and B4 (665 nm): chlorophyll absorbs red and leaves reflect near-infrared, so a green canopy reads about 0.8 and bare soil 0.1–0.2. The estimator (1) reads every clear pixel of the plot window within ±90 days, (2) summarises each input into features (band medians, seven indices, backscatter, canopy

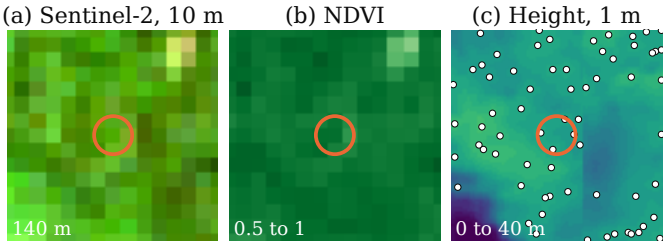


Fig. 2. HCSA plot KT30 (342 Mg ha^{-1}, 23 stems; circle: 0.05 ha): (a) Sentinel-2, 10 m; (b) NDVI, ≈0.9 throughout; (c) 1 m canopy height, E4 tree tops.

height percentiles, GEDI biomass), (3) predicts ln AGB with a model trained on other regions, and (4) converts with $C = 0.47\,\text{AGB}$ and $\text{CO}_2\text{e} = \frac{44}{12}C$.

E4: count the trees, then the formula. E4 rebuilds the right grid's tree list from the 1 m canopy height map, smoothed by a 1 m Gaussian. Tree tops are local maxima above 5 m in a window of radius $0.06H + 0.5$ m [30]; a crown grows from each top while the canopy stays above half its height; and the stem diameter follows from height and crown diameter CD by the tropical-angiosperm model of Jucker *et al.* [31],

$$D = a\,(H \cdot CD)^b \text{ [cm]}, \qquad (a, b) = (0.486,\ 0.802), \quad (4)$$

for Indo-Malaya (0.428, 0.821 Afrotropic; 0.591, 0.781 Neotropic). A satellite cannot identify species, so every tree takes its region's mean tropical ρ in the Global Wood Density Database (0.574 for Southeast Asia) [12], and Eq. 1 gives its biomass. E4 uses no field data. *E4 scaled* makes up for the trees E4 cannot see: it multiplies E4 by the ratio of field to E4 carbon on the other regions.

C. Clouds

For 97 locations we read the Sentinel-2 scene classification of all 22,697 acquisitions of 2021–2023 in a 5×5 window of 20 m pixels (*clear* if every pixel is) and test the scene filter `CLOUDY_PIXEL_PERCENTAGE` < 30 against it. For a share q of target plots we then remove the optical block and compare a naive model, modality dropout [32], a radar model, and radar-imputed optical features.

D. Confidence: Plot and Project Intervals

All intervals target 90% and are calibrated on inner LORO residuals of the training regions. CQR [13] scores $s = \max\{\hat{q}_{\text{lo}}(x) - y,\ y - \hat{q}_{\text{hi}}(x)\}$ on $y = \ln\text{AGB}$ and widens the quantile band by the conformal quantile of s. *Cloud-aware* CQR scores each calibration plot as observed and with its optical block removed, and gives each test plot the threshold of its missingness pattern (CP-MDA [14]); clouds are not independent of the outcome, so coverage must be measured. A project's plot errors share one regional bias, so $SE^2 = s_w^2/n + s_b^2$, with s_w, s_b the within- and between-region spread of log residuals elsewhere. With k field plots, the satellite becomes auxiliary data for a ratio estimator.

E. Conformal Crediting

A registry issues a number, not an interval: a lower bound that should exceed the true stock in at most a share δ of projects [15], [16]. A satellite map has no design-based standard error, so we calibrate the bound on other projects with field truth. Project j has estimate $\hat{Y}_j$, true mean Y_j, relative standard error r_j ($r_j = 1$ if none), and score $z_j = \ln(\hat{Y}_j/Y_j)/r_j$. With m calibration scores, a new project is credited

$$C = \hat{Y}\exp(-q\,r), \qquad q = z_{(\lceil (m+1)(1-\delta)\rceil)}. \quad (5)$$

If projects are exchangeable, $P(C > Y) \le \delta$, as in split conformal prediction with projects as units [8], however plots inside a project are correlated; it needs $m \ge 1/\delta - 1$, nine projects for $\delta = 0.1$. Multiplying the map by c shifts every plain score and q by $\ln c$ and leaves C unchanged: *inflating a map earns no credit*. Regions of 16 or more plots are split into spatial sub-projects of about 10 plots (31 projects in 19 regions), each calibrated only on other regions. We score the over-crediting rate and the *yield*, the mean share C_j/Y_j credited.

F. Ledger and Distributed Extraction

Every protocol step is an entry in an append-only log carrying the SHA-256 of the previous entry [33]: `seal`, `commit` (prediction file digests), `release`, and `reveal`, accepted only if the sealed file still matches; later commits are refused. An extraction task is one (plot, date, window, sensor) request stored under its hash, so re-execution is idempotent and the cache is a checkpoint. Transient errors are retried with capped backoff, and dense tiles are split and placed by consistent hashing [34] or the longest-processing-time (LPT) rule [35].

IV. Results

A. How Certain Is the Right Grid?

On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width, half its length over the plot value, of 31%: the field answer itself is known only to about ±31%, so "100% accuracy" has no meaning. The standards differ too: SNI 7724's root-to-shoot ratio gives 13% more carbon than the 2019 Refinement, and the same Jambi data are deducted 24% (VM0047), 34% (AR-TOOL14), or 14% (Gold Standard). AR-TOOL14 counts sampling error only (9–34% by province), though measurement and allometric error carry 10–64% of the variance of the mean. For growth, VM0047 Eq. 28 with $\rho = 0$ deducts all credit for 13 of 15 SAFE blocks; with the measured ρ (0.45–0.99), seven keep some.

B. Blind Accuracy (RQ1) and Closing the Gap (RQ2)

Table I and Fig. 3a give the blind errors. The confirmatory E2, committed before the reveal, missed by 48.5% overall, 28.6% on the 1 ha units, 41.9% on GEO-TREES, and 102.9% on HCSA, where neither committed estimator beat predicting the mean. The largest gain came from inputs, not algorithms: adding the 1 m canopy height map and GEDI cut the overall error by 9.0 points (bootstrap 95% interval −13.0 to −6.1),

TABLE I
BLIND RRMSE (%) AGAINST THE RIGHT GRID, LEAVE-ONE-REGION-OUT

Left-grid estimate	All	GEO-T.	HCSA	1 ha
E0 Chloris, as published	49.2	37.4	107.1	30.7
E0 GEDI L4A, as published	59.2	50.0	107.0	46.6
E2 (S1/S2, Landsat, PALSAR, DEM)	49.1	42.0	104.8	28.8
+ canopy height + GEDI	40.1	27.7	95.6	20.6
Gain over E2 (points)	−9.0	−14.3	−9.2	−8.2
Embeddings + CH + GEDI	**35.8**	**25.9**	**85.2**	**17.6**

475 plots with AGB $>1\,\mathrm{Mg\,ha^{-1}}$: GEO-TREES 142, HCSA 148 (0.05 ha), Labrière 185 (1 ha). Bootstrap 95% intervals of both gains exclude zero (last row: −4.3, −8.0 to −1.7).

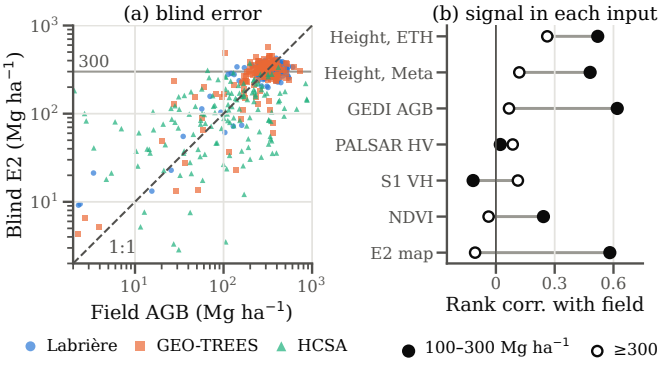


Fig. 3. (a) Blind E2 against field AGB, 475 plots. (b) Each input's rank correlation with field AGB at 100–300 and $\geq$300 $\mathrm{Mg\,ha^{-1}}$.

to 20.6% on the 1 ha units and 27.7% on GEO-TREES, and raised R^2 from 0.29 to 0.53. Read as published, GEDI underestimated by 29–41%: lidar helps as an input, not an answer.

Where the error comes from. *Saturation:* predictions flatten near 300 $\mathrm{Mg\,ha^{-1}}$ (E2 mean 293 above 400, 60 below 50) because the inputs stop changing: from field means of 262 to 480 $\mathrm{Mg\,ha^{-1}}$ the median canopy height rose only from 19.6 to 21.7 m and GEDI from 199 to 207 $\mathrm{Mg\,ha^{-1}}$. Above 300 $\mathrm{Mg\,ha^{-1}}$ no input keeps a rank correlation with the field above 0.26, against up to 0.62 at 100–300, and radar is flat from 100 (Fig. 3b), as reported before [36]; no model can separate what its inputs do not. *Plot size:* the largest of a median 15 stems holds 24% of an HCSA plot's AGB, and one 1 m stem adds about 110 $\mathrm{Mg\,ha^{-1}}$ to a 0.05 ha plot [37].

Aggregation and local plots. With canopy height and GEDI and no field plot, the median error of a block total was 7% on the 1 ha units, 9% on GEO-TREES, and 20% on HCSA; a ratio on $k = 10$–20 local plots (E3) took E2 totals from 29 to 9–10% on GEO-TREES and 11 to 6% on Labrière, but hurt on HCSA until $k = 20$.

C. Counting Trees from Space, and Ten Projects

What the field counts. The median HCSA plot holds 700 stems per hectare of five species, most of them small: stems of 5–15 cm are 75% of the count but 13% of the carbon, and the 6% of 30 cm or more hold 58%. Light wood ($\rho < 0.5\,\mathrm{g\,cm^{-3}}$, mostly Lauraceae and Moraceae) is 24%

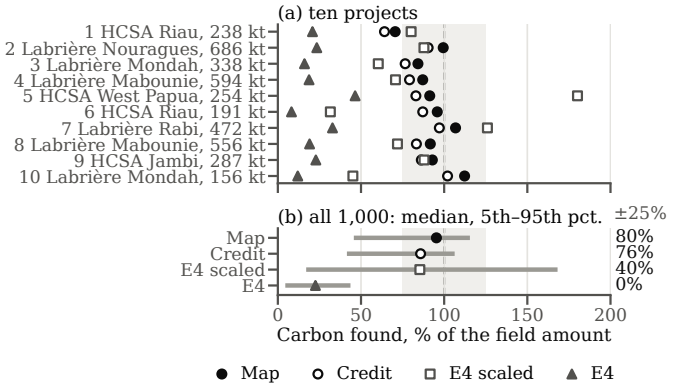


Fig. 4. Carbon found in simulated 1,000 ha projects of 10 random plots, % of the field: (a) the first ten (field kt CO_2e); (b) all 1,000. Map: E2 + CH + GEDI; E4 scaled: E4 × the field-to-E4 ratio of other regions (≈3.8); credit: conformal, $\delta = 0.1$. Shaded: ±25%.

of the stems and 16% of the carbon, medium wood (0.5–0.7, led by Dipterocarpaceae) 44% and 44%, heavy wood (≥ 0.7) 32% and 40%. The regional default ρ instead of each species' value leaves 91% of the total carbon, 78–115% of a plot's (10th–90th percentile).

What the satellite counts. On the HCSA plots E4 found a median of 20 trees per hectare (Fig. 2c): 3% of the stems, 11% of those of 15 cm or more. The map stores whole metres and is smoother than a crown; unsmoothed, every stair step becomes a top (270 per hectare), so the count is a processing choice. E4 found 25% of the field carbon, 23–26% in every dataset; scaled (×3.8) it found 97% of the total, but only 36% of plots within ±25%, against 60% for the map. NDVI (Eq. 3) cannot do better: its median on 136 clear plots was 0.77 below 100 $\mathrm{Mg\,ha^{-1}}$ and 0.83 above 300.

Ten projects. Fig. 4 gives the carbon each grid finds in simulated 1,000 ha projects of 10 random plots of one region. Over all 1,000 the map found a median 95% of the field carbon, within ±25% in 80% of projects; conformal crediting issued a median 86% and exceeded the field in 12%, near its 10% target.

D. Toward Production: Fixes and an Unseen Test

We attacked each limit after the reveal (exploratory, Fig. 5), then froze the method and tested it blind. **Inputs.** The embedding cut plot error from 40.1 to 35.8% (Table I), from Earth Engine alone; the 2026 map added little (35.1%). **Bias.** The map's bias repeats within a realm, so we multiply it by $e^{-\mu}$, μ the mean log map-to-field ratio of the realm's other regions: HCSA projects within ±25% rose from 56 to 68%. Ten local plots added little (1 ha) or noise (0.05 ha). **Credit.** Errors are shared within a region, so one score per region in Eq. 5 cut over-crediting from 13.6 to 9.7%. **Height.** On the 2026 map, 4 m taller, E4 found 62% of the carbon. Our platform's formula $3.836H^{1.131}$ is in no publication, applies 0.47 twice, and found 35%. **Frozen method.** Per 1 ha cell: Earth Engine features, the LightGBM map, the estimate $\hat{Y}e^{-\mu}$, the credit $\hat{Y}e^{-q}$ (q over verified regions), and growth from the IPCC

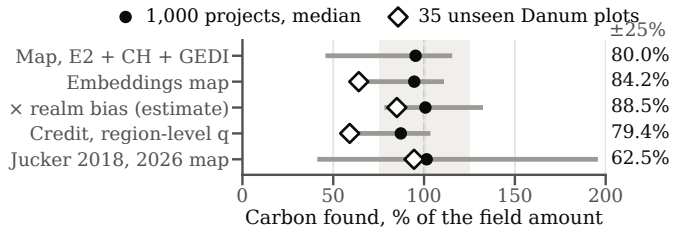


Fig. 5. Each fix on the 1,000 projects (median, 5th–95th percentile; right: share within ±25%) and 35 unseen Danum plots. The credit exceeded the field in 9.7% of projects. Jucker on Danum: post-reveal.

default (0.7 t d.m. ha^{-1} yr^{-1} primary, 2.7–3.4 secondary [3]). **Unseen test.** 35 plots of 0.28 ha measured in 2020 in Danum, Sabah [38] (4,201 stems, 366 Mg ha^{-1}), were sealed before the frozen method, trained without the three GEO-TREES Danum plots, committed. The map found 64% of the field carbon, the estimate 85% (93% of 10-plot draws within ±25%), and the credit 59%, never above the field.

E. Clouds and Confidence (RQ3)

How often is a plot clear? At a median Indonesian location only 21.5% of Sentinel-2 acquisitions see the plot clear (Americas 39%), but revisit compensates: the median gap between clear looks is 10 days. *The scene filter is the real problem*: in Indonesia it keeps 11.5% of acquisitions, 34% of them cloudy over the plot, and discards 65% of the clear ones. Removing every optical feature raised the error only from 48.0 to 50.0%, as the model leans on radar, terrain, and annual products; imputing optical features from radar was harmful (bias −61%).

Which intervals are honest? Only CQR reaches the 90% target with usable width (89.9%, upper bound 5.3 times the lower); normalized conformal under-covers (79.4%), as a difficulty model fitted elsewhere does not transfer. With every target plot clouded, plain CQR falls to 86.5% and cloud-aware CQR keeps 90.5% (+4.0 points, interval +1.9 to +6.6); HCSA stays under-covered (68–81%). A metadata confidence score like our platform's carries no information about error once the dataset is controlled for (partial rank correlation −0.03 to +0.07).

Projects. For 19 projects, intervals assuming independent plot errors covered 74% (HCSA 33%), the two-component interval all 19. As auxiliary data the satellite cut the plots needed for ±10% from 408 to 216 (GEO-TREES) and 381 to 162 (Labrière), none on HCSA.

F. From Intervals to Credits and Growth (RQ4)

We credit the 31 projects at $\delta = 0.10$. Crediting the raw map over-credits 35.5% of projects; VM0047 on the honest two-component uncertainty never over-credits but credits only 48% of the true stock. Conformal crediting meets its target: 9.7% of projects are over-credited, and it credits 72% of the true stock without a field plot, 84% with the canopy height and GEDI map. A better map buys credits, not risk: on five maps, including E2 inflated by 10 and 20%, over-crediting stayed within 2.9 points of the target, and the inflated maps earned the same 72%. With 5 random field plots and a ratio estimator, VM0047 still over-credits 22.6% of projects (yield 82%), 29% with 10, as its deduction starts only above 10% uncertainty; the design lower bound and studentized conformal meet the target (9.9 and 9.6%, yield 81%).

Growth. Excluding oil palm, 196 SAFE plots gained a mean 1.87 Mg ha^{-1} yr^{-1}. The Chloris difference had the error of predicting no change (RMSE 8.06 vs 8.12) and the E2 difference twice that: a 30–100% stock error cannot resolve 1% growth a year. The mean rate of the other blocks was best (block MAE 0.98 vs 2.05 for no change); PALSAR, CCI change, and Hansen loss did not beat it (1.20).

G. Systems (RQ5)

The stock extraction ran 2,052 tasks on 24 workers in 2.1 h; splitting dense tiles cut the worst-worker load factor from 6.75 to 1.40 (LPT), and retries absorbed 85 transient errors. Results were bit-identical to the single-worker run at any worker count, under injected faults, and after a resume; both ledgers verify end to end.

V. Discussion

Can satellites replace the field plot? Not at plot scale, and not for growth. Open data put a 1 ha plot within about 21% (airborne lidar: 13% in Sabah [39]; radar: 31–38% in Kalimantan [40]), a 0.05 ha plot within about 100%. Nor can they count trees: the 1 m map shows a canopy surface, not stems, and the formula applied to it finds a quarter of the carbon, 62% on the 2026 map. Satellites *replace part of the inventory*: project totals, and half the field plots where plots are large.

Confidence must be calibrated, not graded. A metadata score said nothing about error, and the $U-10\%$ deduction is no lower confidence bound; conformal crediting makes the over-crediting rate a design parameter and removes the reward for inflating a map. Its novelty is narrow: the target is Grassi's and Neeff's, the machinery split conformal, and area-level bounds exist [41], but we found no crediting rule calibrated across projects on field truth.

Implications and roadmap. For NusaCarbon: the frozen method of Section IV-D instead of the height formula, plot-level cloud masking, and cloud-aware CQR instead of a letter grade. Stems stay unseen, which motivates drone lidar and under-canopy scans of Indonesian 1 ha plots.

Threats to validity. Only E0, E2, the first change estimates, and the Danum test are confirmatory; all other experiments, E4 included, are exploratory, and we report every variant run. Danum is one valley of tall forest. GEDI and Chloris may have seen some of our sites; HCSA and Labrière dates are known to the year. Projects are region blocks, not registered projects.

VI. Conclusion

Tested blind against a sealed standard-formula right grid, open satellite data reached 21% error on 1 ha plots, 7–9% on project totals, and a median 95% of the field carbon in

1,000 simulated projects, but failed on small plots, tree counts, and growth, and their inputs saturate above $300\,\mathrm{Mg\,ha^{-1}}$. Embeddings and a realm bias correction put 88.5% of projects within $\pm 25\%$, and the frozen method found 85% on unseen Bornean plots. Plot-level cloud masking nearly triples usable optical data, cloud-aware CQR keeps intervals honest, and conformal crediting issues credits at a stated risk without a field plot.

Data and Code Availability

Field data are public [17]–[20]; code, results, the extended version, and a dashboard are at https://github.com/kennytangg/carbon-paper.

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